AI for Good Sandbox Hackathon – PrismAI – Prism: agentic AI approval and assurance for public services

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Github link: https://github.com/devinwarp/prismai_UN_A4G. Licence: AGPL-3.0 (a commercial licence is available from the copyright holders).

1. Introduction

On 23 April 2026 the UAE announced a plan to move 50% of government sectors, services and operations to agentic AI within two years (about April 2028; assumed, as no date was published). On 18 May 2026 the Cabinet adopted a framework to transform at least 50 per cent of federal services and operations into agentic AI systems, and asked each entity to "set targets and establish assessment indicators". The sources we ingested leave three things open: what counts towards the 50 per cent, how much autonomy an agent may be given, and who may approve an agent going live.

Prism is an agentic approval and assurance layer for that gap. A policy is configured (jurisdiction, target %, deadline, counting rule). Use cases are imported and tagged to the four quadrants of the Digital Dubai AI Integration Matrix. Role agents (innovator, policy and legal adviser, security, privacy, compliance, frontline, finance, coordinator) examine each case against a knowledge base built with ITU's reference pipeline. A deterministic policy engine, not a model, issues the verdict, and each entity can set its own human-in-the-loop rules. Every step is written to a hash-chained trace log, and a Sijil panel supplies run-level evidence. A federal dashboard reports deployed % beside compliant %. Two read-only pages (sign-in required), /admin/federal/kb (knowledge-base explorer) and /admin/federal/gaps (the ten gaps of section 6, each chunk id resolved to its quoted text), let a reader check every citation. All entities and data are fictitious.

This report maps the solution to the ITU-T Y.3172 pipeline and to the AI Readiness Report 2.0 dimensions and factors, walks through four staged error scenarios, and lists the policy and standards gaps the mapping exposed.

2. Description of the use case

Domain: public services. Primary case UC02, trade-licence renewal and permit pre-check at Marsa Public Services Authority (fictitious): an external agent pre-checks a renewal and drafts the customer message. It sits in the Matrix's External Agents quadrant and is client-facing, so the entity rule is that an officer gives final approval before any message reaches a customer. The federal view covers 20 use cases across five fictitious entities (8 internal agents, 5 external agents, 3 internal RAG, 4 external RAG).

Gap in existing solutions. Among the sources we ingested, governance frameworks and readiness tools state principles and Y.3172 describes pipelines, but none gives a government a testable, role-owned and tamper-evident way to decide whether an agent may go live, or reports adoption speed beside verified compliance. Prism supplies that decision layer.

- **Counting rule.** The 50% target has no published metric. Prism makes the rule explicit and editable (default: a use case counts when its verdict is GO_AUTONOMOUS or GO_SUPERVISED) and reports compliant % beside deployed %, so speed cannot hide a safety failure.
- **Autonomy levels.** A five-step verdict scale (GO_AUTONOMOUS, GO_SUPERVISED, ASSISTANT_ONLY, NO_GO, ESCALATE) with a hard cap: a client-facing use case never exceeds GO_SUPERVISED.
- **Human in the loop.** Per-entity rules, edited on the Federal page (src/lib/un/hitl.ts): client-facing output needs a person's final approval; decisions affecting rights or fees need a human decision; irreversible actions need human approval. All three default to on, an entity can switch one off, and an applicable rule declared unmet is a hard failure of the human-approval gate (verdict NO_GO), never averaged. The rules load into runs from the entity configuration.
- **Accountability.** Each hard gate has one owning role agent; only the owner can fail or pass it, and only the applicant can declare a fix. The verdict is computed by code from gate states, never averaged.
- **Auditability.** Every agent message, every rejected message and every verdict is a hash-chained record with quoted citations, and a run can be replayed from its log. Each trace record is hashed together with the previous record's hash, and the hash covers every field of the record, including its timestamp, the exact agent message, the evidence and knowledge-base citations, the proposal and disposition, the trace ids a verdict rests on, any HITL breach and the human signature. Editing any record, or deleting, inserting or reordering records, breaks the chain. A reader can check it offline with verify_log, which names the first broken record; dropping the last records is detected by

comparing with the stored head hash. Separately, every change to a use case's stage, verdict or compliant flag, to the policy and to the human-oversight rules is written to an append-only, hash-chained change log with the old and new value and a UTC timestamp, verified by a database function that names the first broken row. The change log records the database role, not the person; who signed a verdict is on the use case and in the run trace. History starts when the log was installed.

3. Use case requirement

Requirements per Y.3172 node. Automated tests cover the policy, gate, agent, trace and scenario code (R1 and R3 to R7): on a fresh clone, npm ci && npm test gives 122 tests, 121 pass and 1 is skipped. R2 is checked by running the knowledge-base build.

SRC. Every knowledge-base source carries its URL, a SHA-256 and, where the source states one, its date (22 sources, 331 chunks). Copyrighted sources (UAE AI Charter, UAE AI Strategy 2031, the PDPL official text, Dubai AI Ethics Guidelines) are held as short attributed excerpts with URLs, not full text. A 20-use-case CSV and staged scenario events are the input; events are released one step at a time and unevenly distributed, never all at once (R7). Sijil telemetry is simulated unless the optional live adapter is used. Synthetic data only.

C. The knowledge base is rebuilt by one script (kb/build.py) using ITU's parse, chunk and embed functions; the runtime needs no Python. CSV import reports errors by line. The hosted app needs approved infrastructure for any agent it assesses (the infrastructure gate).

PP. Chunks carry jurisdiction, instrument type, binding flag, date and Matrix quadrant; CSV rows are validated; Sijil records are grouped into runs. Chunks are vectorised with ITU's embedding function and model (nomic-embed-text, 768 dimensions, kb/vectors.json).

M. At least 5 role agents are active (8 are defined); each sees only the events routed to it and speaks in typed messages (PROPOSE, CHALLENGE, EVIDENCE_REQUEST, EVIDENCE, POSITION_UPDATE, ESCALATE, VERDICT) (R3). A query returns cited chunks from UAE/Dubai and global sources (R2). The app searches by keyword (BM25); hybrid keyword-plus-vector search is built but switched off in the app.

P. Five hard gates (approved infrastructure, audit log, lawful data, human final approval, AI disclosure); optional metric gates are added by a scenario (hallucination rate and fee accuracy in S2; no commercial promotion and purpose limitation in S4). A message with no evidence, or citing a chunk that does not exist, is rejected (R4). Per-entity HITL rules and the client-facing cap apply (R1 configuration for target, deadline and counting rule).

D. The dashboard reports deployed % and compliant % separately (R1); a run viewer shows the step timeline, gate matrix and hash-chain status; pilot reports are filed as hashed Markdown.

SINK. Verdicts list the trace records they rest on, and the verifier detects edits, deletions, insertions and reordering (R5). Evidence per use case: at least 30 Sijil runs with accuracy ≥ 95%, hallucination ≤ 1%, goal alignment ≥ 90% and zero rejects on hard criteria (R6; the three rates are simulated here).

4. Mapping with ITU-T Y.3172, the AI Readiness Report 2.0, and domain-specific policies

Report 2.0 factors: Data (knowledge base, use cases); Research (Singapore, EU and NIST benchmarks); Deployment support (infrastructure gate); Standards (Y.3172 pipeline, Matrix quadrants, verdict scale); Open source and code (GitHub; ITU parsing and chunking functions reused; HTML cleaning is ours); Sandbox environments (scenario engine, simulated Sijil).

Node	Prism use case	Readiness 2.0 dimension	Documents in the knowledge base (policies, strategies, standards)
SRC	22 sources (331 chunks) pinned by SHA-256; 20-use-case CSV; staged scenario events; Sijil telemetry (simulated)	1 Data/Model Marketplace; 4 Contextualization and Regional Impact	UAE Cabinet, 23 April and 18 May 2026 (50% target); Digital Dubai AI Integration Matrix; UAE AI Strategy 2031; Dubai AI Ethics 2019 (excerpts)
C	ITU reference ingestion code (parse, chunk, embed); CSV import with line-level errors; simulated and optional live Sijil adapter	13 Digital Infrastructure	National Cyber Security Policy for AI (UAE); DESC Information Security Regulation v3.1 (excerpts; cloud security controls, desc-isr-excerpts#0); Recommendation ITU-T Y.3172; ITU AI Readiness reference repository
PP	Chunks tagged with jurisdiction, instrument type, binding flag, date, Matrix quadrant; CSV validation; Sijil run grouping	3 Cross-Domain Correlation Analysis	Federal Decree-Law 45 of 2021 (PDPL): lawful processing and purpose limitation; ITU AI Readiness 2025 on standard data formats and personal-data standards (itu-ai-ready-2025#27); ITU chunking code; kb/sources.json schema
M	Eight role agents (model-driven mode built, deterministic mode evaluated); BM25 retrieval. Agents draft customer messages (UC02) and answer citizens (UC01, UC13), so the hallucination-rate and fee-accuracy gates act as the generated-content control	2 Generated Content Marketplace; 5 Level of Integration of AI in Workflows; 6 Human Interface; 8 Collaboration with AI	ITU AI Readiness 2025 on trustworthy generative AI (itu-ai-ready-2025#54); AI Integration Matrix (internal/external, agent/RAG); UAE AI Charter (human-machine ties); NIST AI RMF 1.0

Node	Prism use case	Readiness 2.0 dimension	Documents in the knowledge base (policies, strategies, standards)
P	Policy engine, the Y.3172 P node: five hard gates, per-entity HITL rules, verdict scale, client-facing cap, citation gate	10 AI & Policies; 7 Strategy Alignment	UAE AI Charter (human oversight, transparency); Singapore agentic AI framework (law-firm summary); EU AI Act Annex III (access to essential public services)
D	Federal dashboard; trace viewer; KB explorer and gaps pages; auto-filed pilot reports (hashed)	7 Strategy Alignment; 12 Granular Priorities	UAE Cabinet (entity targets and assessment indicators). Energy: none ingested (see note)
SINK	Hash-chained trace log and filed reports; compliant flag read by the dashboard; human signature on the verdict	9 Impacts of Humans in AI Integration; 10 AI & Policies	UAE AI Charter (accountability); NIST AI RMF (govern, measure)
MLFO / intent	Coordinator agent and scenario engine; the policy configuration and each use case's requested autonomy are the intent	7 Strategy Alignment	UAE Cabinet (targets and indicators per entity); Singapore framework (lifecycle accountability)
ML sandbox	Scenarios S1 to S4 and the simulated Sijil adapter run on synthetic data only; S3 works on a sandbox copy	9, 10 (sandbox factor); 11 AI for Inclusion (partial)	Dubai Future Foundation AI Sandbox and Agentic AI accelerator; ITU AI4G sandbox summary; Dubai AI Ethics 2019 and UAE AI Charter (fairness, privacy) for dimension 11

Dimension 11 (AI for Inclusion), partial. Covered: citizens are told they speak to an AI (disclosure gate), a person approves client-facing output, and S4 blocks profiling on inferred sensitive attributes (including disability). Not done: no accessibility, language or digital-divide testing, and no ingested policy on inclusive access to agentic services. **Energy and sustainability.** No energy or sustainability policy was ingested, and we did not measure Prism's compute or energy use. The ITU material in the knowledge base only notes that Y.3172 addresses energy characteristics of pipeline nodes (itu-y3172-summary#3) and that large models consume increasing energy (itu-ai-ready-2025#54).

5. Error scenarios.

S1, UC02 (gate engine). Each event reaches only the roles named; verdict is computed by code.

1. Step-1 (T0): Marsa requests GO_AUTONOMOUS for a client-facing case. Prism: baseline results go to the innovator, frontline and policy adviser; audit-log, lawful-data and disclosure evidence go to security, privacy and compliance; hosting and human-approval gates have no evidence. ESCALATE (autonomy above the client-facing cap; two gates unevidenced).
2. Step-2 (T1): only the security agent learns the model endpoint is not on DESC-certified infrastructure. Prism: CHALLENGE citing desc-isr-excerpts#0; NO_GO. No other agent sees the event.
3. Step-3 (T2): only the compliance agent learns the agent sends customer messages without officer release. Prism: second gate fails, citing the Singapore framework (law-firm summary): "human oversight mechanisms that can effectively override, intercept or review agentic AI actions." NO_GO.
4. Step-4 (T3): the innovator declares both fixes. Prism: declared but untested, so ASSISTANT_ONLY (the agent prepares, a person acts).
5. Step-5 (T4): re-tests pass. Prism: GO_SUPERVISED, the ceiling for a client-facing case. The path matched the expected path at every step (ESCALATE, NO_GO, NO_GO, ASSISTANT_ONLY, GO_SUPERVISED); the run wrote 24 trace records from 7 active agents; the verifier passes and changing one claim in record 4 fails verification at record 4.

S2, hallucination alert, UC01 voice contact centre (GO_SUPERVISED pilot, client-facing). Sijil figures are simulated scenario inputs.

1. Step-1 (T0): pilot live, officer approves every written follow-up. Prism: all five gates pass; GO_SUPERVISED.
2. Step-2 (T1): only the compliance agent sees a Sijil alert (hallucination 6.1% over 3 runs against a 1% threshold); the owner restricts fee answers to the fee table. Prism: fix declared, not re-tested; ASSISTANT_ONLY.
3. Step-3 (T2): only the frontline agent sees a complaint that the assistant quoted a licence fee that does not exist. Prism: compliance links it to the T1 alert and adds the fee_accuracy gate, which fails; NO_GO.
4. Step-4 (T3): only the policy adviser sees the requirement to tell users when AI-generated information may be inaccurate. Prism: the disclosure gate is re-tested against it and fails; two open failures are escalated to a person; NO_GO.
5. Step-5 (T4): the owner declares fixes for both gates. Prism: ASSISTANT_ONLY.
6. Step-6 (T5): all re-tests pass (hallucination 0.6% over 3 runs). Prism: GO_SUPERVISED, with the hallucination and fee-accuracy gates now permanent. Expected path: GO_SUPERVISED, ASSISTANT_ONLY, NO_GO, NO_GO, ASSISTANT_ONLY, GO_SUPERVISED.

S3, federal cascade (sandbox portfolio simulation). Policy 50% over the 20 use cases, run on a sandbox copy; it is not written to the use-case rows.

1. Step-1 (T0): policy configured. Prism: 0 of 20 deployed (target 10, gap 10).

2. Step-2 (T1): entities report 6 use cases deployed (UC02, UC04, UC06, UC10, UC14, UC19). Prism: 6 of 20 deployed (30%, gap 4), 0 compliant.
3. Step-3 (T2): a Sijil attestation fails 2 of the 6 (UC02, UC06). Prism: 4 of 20 compliant (20%), 6 of 20 deployed (30%).
4. Step-4 (T3, simulated policy update): a new Phase One service category (transport and mobility) is added. Prism: the four transport use cases (UC13 to UC16) are flagged for re-scoring; figures unchanged. Reporting both percentages is the point.

   Not the live app. The scenario figures (6 of 20 deployed, 4 of 20 compliant) are the simulation. The live seeded dashboard reads 2 of 20 deployed and 2 of 20 compliant (UC01 and UC02, signed by a person and attested) against a target of 10 use cases: 10%, a gap of 8. The target is not met and Prism reports that.

S4, UC13 citizen mobility and fines assistant (Qanat Transport and Mobility Agency; the judges' added scenario): a consumer complaint escalates to a privacy and ethics breach.

1. Step-1 (T0): the pilot is live and the entity requests GO_AUTONOMOUS for a client-facing assistant; hosting, audit-log, lawful-data and disclosure gates pass, human final approval has no evidence. Prism: ESCALATE.
2. Step-2 (T1): only compliance and frontline learn that citizens complain the assistant promotes paid third-party parking and insurance add-ons; the chat logs confirm it. Prism: consumer-protection case; the no_commercial_promotion gate is added and fails; NO_GO.
3. Step-3 (T2): only security and privacy see the privacy-log review: the promotions are targeted with attributes inferred from fines and permit data, such as financial hardship and a disability inferred from accessible-parking permits. Prism: this is now a data-privacy and ethics breach (data used for advertising, incompatible with its declared purpose, through profiling); the purpose_limitation gate is added and fails; two open failures are referred to human oversight; NO_GO.
4. Step-4 (T3): only the policy adviser surfaces the PDPL (specific and clear purpose, objection to profiling, human review), Dubai AI Ethics and UAE AI Charter text. Prism: stronger guardrails become permanent for the entity: no inferred sensitive attributes for any commercial purpose, and referral to human oversight; NO_GO.
5. Step-5 (T4): the owner removes the promotions module and blocks inferred-attribute use. Prism: declared but untested; ASSISTANT_ONLY.
6. Step-6 (T5): re-tests pass (no promotions in 60 test chats; data used only for its declared purpose) and an officer reviews every referred case. Prism: GO_SUPERVISED, the ceiling for a client-facing case. The full path equals the declared expected path (6 of 6 steps), checked by an automated test.

   Controversy not solved by Prism. Prism makes the guardrails permanent for the entity and refers the case to the entity's own human oversight. It has no workflow that refers a matter to a national AI ethics body, and it imposes no sanction or remediation; that remains a human and legal step outside the tool.

6. Policy gaps and standards gaps derived from the mapping exercise

Domain-level gaps in governing agentic AI for public services (kb/gaps.json, G1 to G10; shown on /admin/federal/gaps), not recommendations to any country. UAE and Dubai instruments are the reference implementation. Every chunk id resolves in kb/chunks.json. Grounding is weak where marked.

Gap	Ingested national sources	Global example	Prism response (and caveat)
G1 Responsibilities: no instrument says which role answers for an agent's outcome across entities	Roles per entity are defined but not who answers for an agent action (uae-cabinet-2026-05-18#3); accountability split between designers, developers and deployers without naming who decides (dubai-ai-ethics-2019#5)	NIST AI RMF: responsibilities by lifecycle role; roles must be defined and differentiated (nist-ai-rmf#17, #66)	One owning role per gate. Caveat: national evidence is an absence of text in short excerpts.
G2 Responsibilities: who is the overseeing person, with what authority, and what they sign	Oversight stated as a principle (uae-ai-charter#2); override by designated people, designation open (dubai-ai-ethics-2019#10)	NIST MAP 3.5 (nist-ai-rmf#46); Singapore agentic framework (sg-agentic-summary#2, #3) and EU summary (itu-eu-ai-act-summary#0), both summaries	Human final-approval gate, HITL rules, recorded human signature. Weak grounding: two of three examples are secondary summaries; EU Art. 14 not ingested.
G3 Standards: no shared, testable standard for fit to go live	Self-assessment tool (dubai-ai-ethics-2019#13); proposed quality mark (uae-ai-strategy-2031#7)	NIST MANAGE 2.4, no numeric threshold (nist-ai-rmf#54); Y.3172 ML sandbox, for networks (itu-y3172-standard#35)	Deterministic thresholds, hard criteria never averaged. Caveat: thresholds are demo defaults, not from a standard; attestation simulated.

Gap	Ingested national sources	Global example	Prism response (and caveat)
G4 Standards: risk classes differ in kind, so an agent cannot be classified once and reused	Four Matrix quadrants (dubai-ai-matrix-launch#2)	EU AI Act Annex III lists high-risk areas, incl. public benefits (eu-ai-act-annex3#0, #3); four tiers (itu-eu-ai-act-summary#0)	Three observable facts (client-facing, affects rights, irreversible) map to gates. Caveat: a simplification of Annex III.
G5 Technology: bounded tool access and approved hosting stated at requirement level	Security tailored to the AI application (uae-cyber-ai-policy#3); cloud controls by the provider (desc-isr-excerpts#0)	Singapore: control tool access, least privilege (sg-agentic-summary#4, #5)	Infrastructure gate; agents see only released events. Weak grounding: one global example, a law-firm summary.
G6 Technology: no tamper-evident audit-record format specified	Audit trail with complete logging of agent actions (dubai-ai-matrix-excerpts#0); AI monitoring (uae-cyber-ai-policy#2)	NIST MANAGE: incident response, no record format (nist-ai-rmf#54)	Hash-chained trace and audit-log gate. Weak grounding: the NIST chunk shows incident handling exists, not that logging is unspecified.
G7 Legislative: personal-data law gives rights against automated decisions but excludes government entities	PDPL does not apply to government data or entities (uae-data-protection-laws#2); objection and human review (#10, #11)	EU Annex III places public-benefit decisions among high-risk systems (eu-ai-act-annex3#3), a contrast, not a data law	PDPL-style rights applied as policy (lawful-data gate, human decision on rights or fees). Weak grounding: one excerpt; not legal advice.
G8 Legislative: most agentic-AI instruments are non-binding	Charter is a non-binding framework in our metadata (uae-ai-charter#3); regulation as a strategy aim (uae-ai-strategy-2031#5)	EU Annex III, binding contrast (eu-ai-act-annex3#0); UN dialogue news report (un-global-dialogue#1)	Gates give a non-binding rule a testable pass or fail. Weak grounding: the UN item is a news report, not a UN instrument.
G9 Coordination: no stated way to reconcile federal, city and sector rules	Collect once, use across entities (uae-cabinet-agentic-50#4); multi-body Dubai accelerator (dff-agentic-accelerator#2)	ITU readiness: fragmented governance limits participation (itu-ai-ready-2025#14); UN dialogue (un-global-dialogue#2)	One dashboard, same gates and trace for every entity. Weak grounding: the ITU item is general; the UN item is a news report.
G10 Conflicting objectives: adoption speed versus assurance	Entities assessed on adoption (uae-cabinet-agentic-50#1, #2) while safety is a stated priority (uae-ai-charter#0)	NIST: unrealistic risk expectations can misallocate resources (nist-ai-rmf#15)	A failed hard gate rules out every GO verdict; deployed % beside compliant %. Weak grounding: the tension is inferred; no source says they conflict.

Sources not ingested. No G7 document and no UN instrument: the one UN item is a news report (un-global-dialogue). Singapore's framework PDF, EU AI Act Articles 14 and 86 and the full PDPL text are not in the knowledge base; the Singapore items are a law-firm summary. **Limits.** Sijil is simulated by default and labelled so. The optional live adapter takes run counts, policy decisions, approval records and the signed hash-chained ledger verification from a Sijil node; it is tested only with a mocked fetch, not against a real node. Sijil does not measure accuracy, hallucination or goal alignment, so these stay simulated. Agents were evaluated in deterministic mode; model-driven mode is built but not yet scored. Retrieval is lexical, so citation relevance is uneven. Retrieval was measured on 30 questions over the 331-chunk, 22-source knowledge base (kb/eval/RESULTS.md). The keyword search the app uses (BM25) reaches recall@5 of 0.87, hybrid keyword-plus-vector search 0.83 and vector-only search 0.50; on 14 held-out questions the figures are 0.86, 1.00 and 0.71. The plan's hybrid target of 0.85 is not met. Vector search is switched off in the live app and BM25 is the default; we do not claim vector search is live. The excerpt-chunk vectors are approximate, not the model's output, so vector and hybrid figures are indicative.